

An Approach to a Case of Wheezing: Asthma

Dr. Essam Kotb, Associate Professor, Internal Medicine | Source: 5-Asthma_updated.pdf (34 slides) · Refs: Step-Up 5th ed.; Harrison 20th ed.; AMBOSS; Cecil 10th ed.; First Aid IM Boards 4th ed.

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from the COPD lecture (same lecturer):** COPD = **irreversible** obstruction in older **smokers**, driven by **neutrophils and proteases**. Asthma = **reversible** obstruction, often starting in **childhood**, driven by **Th2 cells and eosinophils**. The objectives explicitly ask you to **differentiate the two**, so the comparison diagram comes first.

Asthma vs COPD: the comparison examiners love

Feature	ASTHMA	COPD
Onset	Usually < 35 y, often childhood	Usually > 50 y
Smoking	Rarely smokes	Smoker / ex-smoker (~90%)
Symptoms	Episodic, variable day to day	Persistent, progressive dyspnoea
Cough	Dry, wheeze, worse at night	Chronic productive, worse in morning
Inflammation	Th2, eosinophils, mast cells (type I)	Neutrophils, proteases
Spirometry	FEV1/FVC < 0.7, REVERSIBLE	FEV1/FVC < 0.7, NOT reversible
Reversibility	> 12% ↑ FEV1 after bronchodilator	Stays < 0.7
DLCO [added]	Normal or ↑	↓ in emphysema
First controller	ICS from step 2 (never LABA alone)	LAMA (ICS only if eos ≥ 300 / admitted)
Acute O₂ target	93–95% (children 94–98%)	88–92%
Bad ABG sign	NORMAL or ↑ PaCO ₂ (tiring)	Acidosis: pH ≤ 7.35 + PaCO ₂ ≥ 45 → NIPPV

Added diagram: asthma vs COPD side by side

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Define asthma (chronic inflammation, hyperresponsiveness , reversible obstruction)	Definition keywords
Knowledge	★ Allergic vs non-allergic; clinical subtypes	Extrinsic vs intrinsic table; aspirin, exercise, cough-variant
Knowledge	★ Th2 inflammation, type I hypersensitivity, mast cells, smooth-muscle hypertrophy, oedema, mucus plugging	Mechanism MCQ
Knowledge	Symptoms and signs	Silent chest, pulsus paradoxus
Knowledge	★ Diagnostic tests; severity classification (mild intermittent → severe persistent)	Spirometry reversibility > 12%; methacholine; classification table
Knowledge	★ Controller/reliever drugs; acute exacerbation protocol	Stepwise steps 1–6; acute algorithm
Skills	Interpret history, exam, spirometry, PEF, ABG, CXR; classify severity; plan treatment; differentiate from COPD	Case vignettes

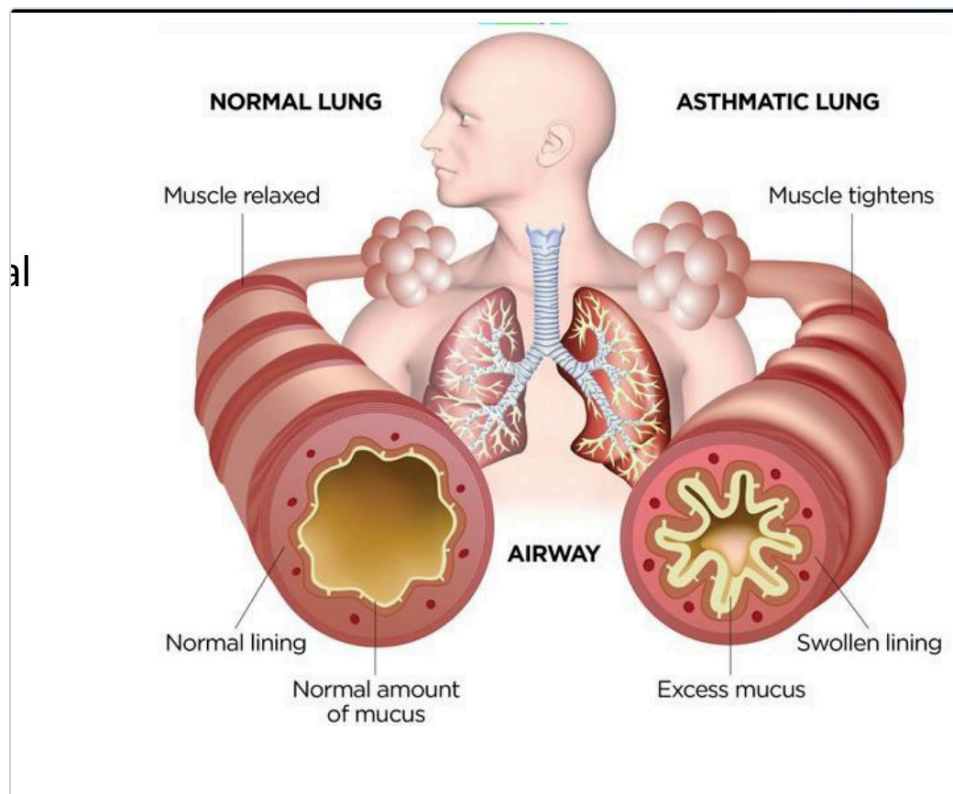
Domain	Objective	Likely tested as
Values	Inhaler technique, adherence, trigger avoidance, self-monitoring	

Flow: Definition → Epidemiology & aetiology → Pathophysiology → Clinical features → Subtypes → Diagnostics → Classification → Management → Acute exacerbation (definition, risk factors, pathogenesis, features, management) → Summary → Cases.

2. Main Content (Lecture Order)

2.1 Definition (Slide 5)

★ **Asthma = a chronic inflammatory disease** of the respiratory system characterised by **bronchial hyperresponsiveness**, causing respiratory symptoms and a ★ **reversible airflow obstruction**.



Slide 5: Normal vs asthmatic airway: tightened muscle, swollen lining, excess mucus

2.2 Epidemiology & Aetiology (Slides 6-7)

Type	Typical onset
★ Allergic	Childhood
★ Non-allergic	> 40 years

Risk factors: ★ **family history** · **allergy** · ★ **atopy** · environmental and socioeconomic factors (**pollution from cooking and heating**).

★★ **Allergic vs non-allergic (Slide 7 table):**

	★ Allergic	★ Non-allergic
Synonym	Extrinsic	Intrinsic
Onset	Early	Late
Worse with allergens	+	–
Personal / family atopy	+ / +	– / –
★ NSAID intolerance	Less frequent	★ More frequent
Pathophysiology	★ Type I hypersensitivity	Unknown
Total IgE	★ Elevated	Normal
Specific IgE	+	–
Eosinophils (blood/sputum)	+	+ or –
Course	More favourable	Less favourable

★ Slide 7 highlight (red): **non-allergic asthma is "irritant asthma", mostly with aspirin or NSAIDs.**

CHARACTERISTIC	ALLERGIC ASTHMA	NONALLERGIC ASTHMA
Synonym	Extrinsic asthma	Intrinsic asthma
Onset	Early	Late
Worsening with allergens	+	–
Personal history of atopy	+	–
Family history of atopy	+	–
Intolerance to NSAIDs	Less frequent	More frequent
Pathophysiology	Type I hypersensitivity	Unknown
Total IgE	Elevated	Normal
Specific IgE	+	–
Eosinophils (blood/sputum)	+	+ or –
Evolution	More favorable	Less favorable

Slide 7: Allergic vs non-allergic asthma

🔄 **Recap: Who.** Allergic (extrinsic) = childhood, atopy, high IgE, type I, better outlook. Non-allergic (intrinsic) = after 40, normal IgE, often aspirin/NSAID-triggered, worse outlook.

🔗 **Link:** Both share the same three airway processes.

📌 **Up next: Pathophysiology.**

2.3 Pathophysiology (Slide 8)

★ An inflammatory disease driven by ★ **T-helper type 2 (Th2) cells** in **genetically predisposed** people. Three processes:

1. ★ **Bronchial hyperresponsiveness**

2. ★ **Bronchial inflammation**: inhaled antigen → **eosinophil activation** → **submucosal oedema** → **bronchiolar collapse**
3. ★ **Endobronchial obstruction** from:
 - ↑ **parasympathetic tone** → **reversible bronchospasm**
 - ↑ **mucus production**
 - **Mucosal oedema with goblet-cell hyperplasia**

[added from the objectives, not on the slide: allergen cross-links IgE on mast cells → degranulation (histamine, leukotrienes) → early-phase bronchospasm; Th2 cytokines (IL-4, IL-5, IL-13) drive IgE, eosinophils and mucus → late phase; chronic remodelling = smooth-muscle hypertrophy.]

🔄 **Recap: Mechanism.** Th2-driven hyperresponsiveness + eosinophilic inflammation + obstruction (bronchospasm, mucus, oedema/goblet cells). All largely reversible.

🔗 **Link:** These processes come and go, which is why symptoms are episodic.

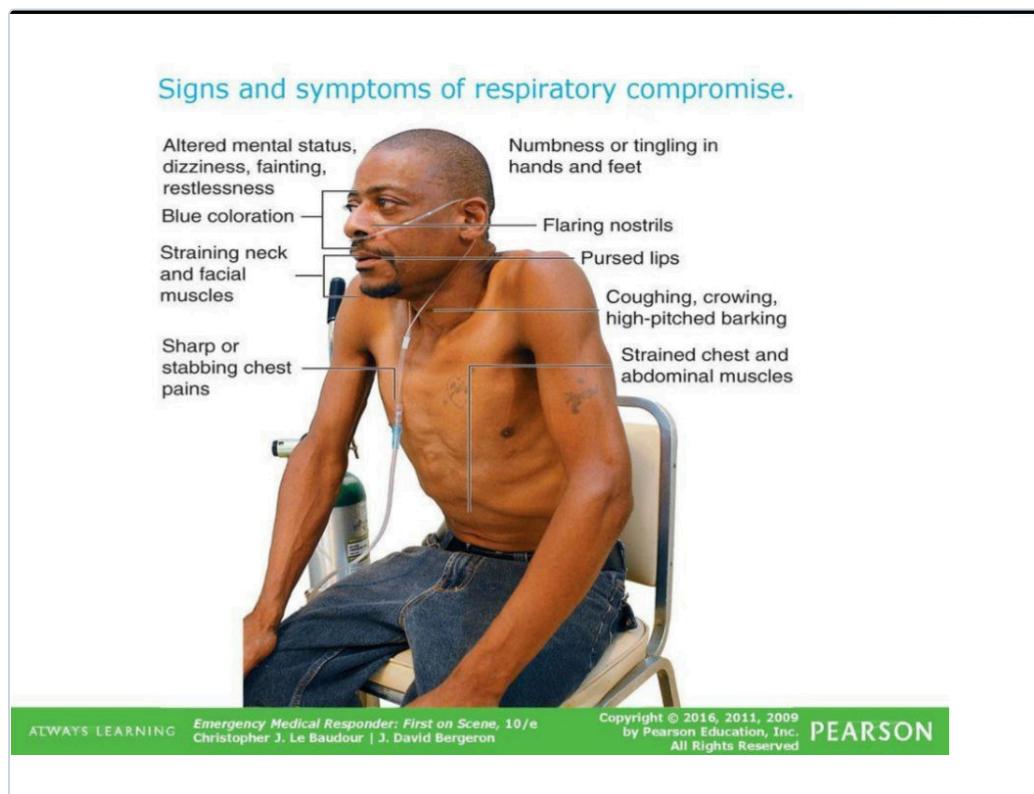
▶ **Up next: Clinical features.**

2.4 Clinical Features (Slides 9-10)

History:

- Symptoms triggered by stimuli (e.g., **dust**); ★ **usually asymptomatic between attacks**
- During attacks: ★ **dyspnoea**, ★ **cough** (persistent dry; ↑ with exercise and irritants; ★ **worse at night**), **chest tightness**, ★ **wheezing**
- **Past history:** ★ **atopy** (eczema/atopic dermatitis, allergic rhinitis, allergic conjunctivitis)
- **Family/social:** atopy or asthma; **mould, carpet, tobacco, pets**

Signs: wheeze · prolonged expiration · ★ **silent chest (severe)** · hyperresonance · tachypnoea, tachycardia · **cyanosis** · **lethargy** · ★ **pulsus paradoxus**



Slide 10: Signs of respiratory compromise

2.5 Subtypes & Variants (Slides 11-13)

Subtype	Key features
★ Allergic	Most common ; childhood intermittent symptoms; allergen-triggered; atopy ; ★ responds well to ICS
★ Non-allergic	Less common; triggered by viral URTI, cold air, GERD ; no atopy; ★ responds poorly to ICS
Adult-onset	Uncommon; first symptoms in adulthood; usually non-allergic; poor ICS response
★ Cough-variant	Chronic dry cough without other typical asthma symptoms
Aspirin-exacerbated respiratory disease	<i>[added: asthma + nasal polyps + aspirin/NSAID sensitivity (Samter triad)]</i>
Asthma-COPD overlap	Features of both
Occupational	Workplace exposure
★ Exercise-induced	Symptoms within 15 min of intense exercise, resolve within 1 h

★ Exercise-induced asthma management:

- Non-pharmacological measures
- ★ **Inhaled β -agonist 15-30 min before exercise**
- ★ **Needed ≥ 4 times/week $\rightarrow$ daily ICS**

🔄 **Recap: Subtypes.** Allergic (common, ICS-responsive) vs non-allergic/adult-onset (poor ICS response). Cough-variant = cough only. Exercise-induced: onset within 15 min, gone in 1 h; SABA 15-30 min before; daily ICS if $\geq 4\times/\text{week}$.

🔗 **Link:** Suspicion needs objective proof: obstruction that reverses.

📺 **Up next: Diagnostics.**

2.6 Diagnostics (Slides 14-18)

★★★★ Spirometry (PFT):

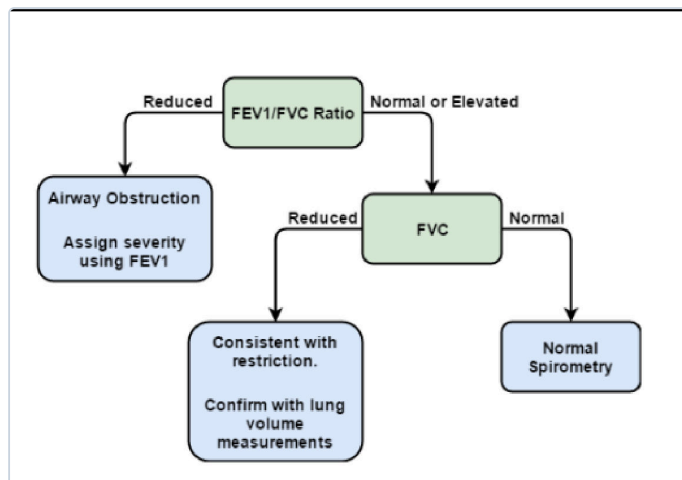
- ★ **FEV1/FVC $< 70\%$** (obstructive), **$\downarrow$ FEV1**
- ★ **Reversibility: $> 12\%$ $\uparrow$ in FEV1 after bronchodilator**

★★ **Bronchoprovocation test:** ★ **$\downarrow$ FEV1 $> 20\%$ with methacholine or histamine** *[added: used when spirometry is normal but asthma is suspected]*

Peak expiratory flow (PEF): varies with **age, sex, height**; normal $\approx 400\text{-}600$ L/min

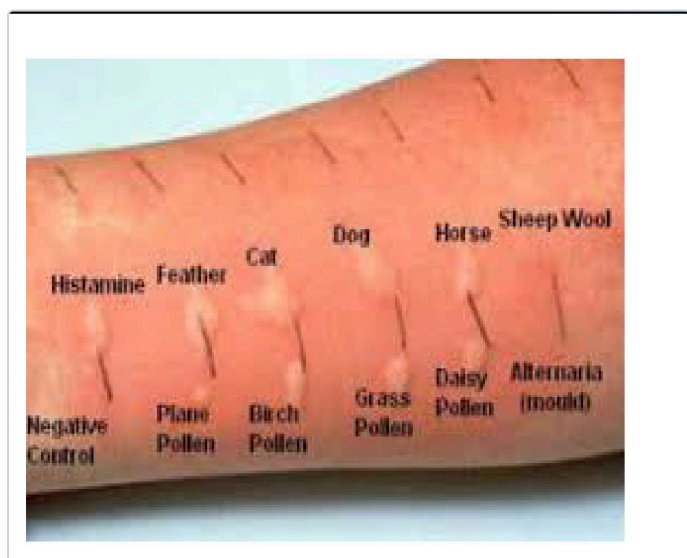


Slide 14: Peak flow meter

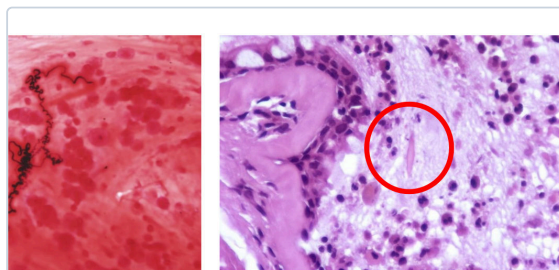


Slide 15: Spirometry interpretation algorithm

Test	Findings
CBC	Possible ★ eosinophilia ($\pm$ leukocytosis)
Antibodies	★ $\uparrow$ total IgE , $\uparrow$ allergen-specific IgE
Skin allergy tests	Skin prick or intradermal
★ Sputum	May look purulent; ★ Curschmann spirals (mucus plugs of shed epithelium); ★ Charcot-Leyden crystals (eosinophilic inflammation)
CXR	★ Normal in mild-moderate ; hyperinflation in severe; needed in severe asthma to exclude other conditions
★★ ABG	★ Hypocarbica is common ; ★ normal or $\uparrow$ PaCO₂ is a BAD sign (bad prognosis if hypercapnic)



Slide 16: Skin prick test



Slide 17: Curschmann spiral and Charcot-Leyden crystals

[added: why a normal PaCO₂ is bad: an attack causes hyperventilation, so CO₂ should be low; a "normal" or rising CO₂ means the patient is tiring.]

🔗 **Recap: Diagnosis.** Obstruction (< 70%) that reverses (> 12% FEV1). If spirometry is normal: methacholine (> 20% fall). Eosinophilia, ↑ IgE, positive skin prick, Curschmann spirals, Charcot-Leyden crystals. CXR often normal. In an attack, low CO₂ is expected; normal/high CO₂ is danger.

🔗 **Link:** Once confirmed, severity classification decides the starting treatment step.

📌 **Up next: Classification and stepwise management.**

2.7 Classification & Stepwise Management (Slides 19-21)

☆☆☆ Severity classification:

Class	Daytime symptoms	Night-time symptoms	PEF or FEV1
Mild intermittent	≤ 2/week	< 2/month	≥ 80%
Mild persistent	> 2/week but < 1/day	> 2/month	≥ 80%
Moderate persistent	Daily	> 1/week	60-80%
Severe persistent	Continuous	Frequent	< 60%

Asthma Classification

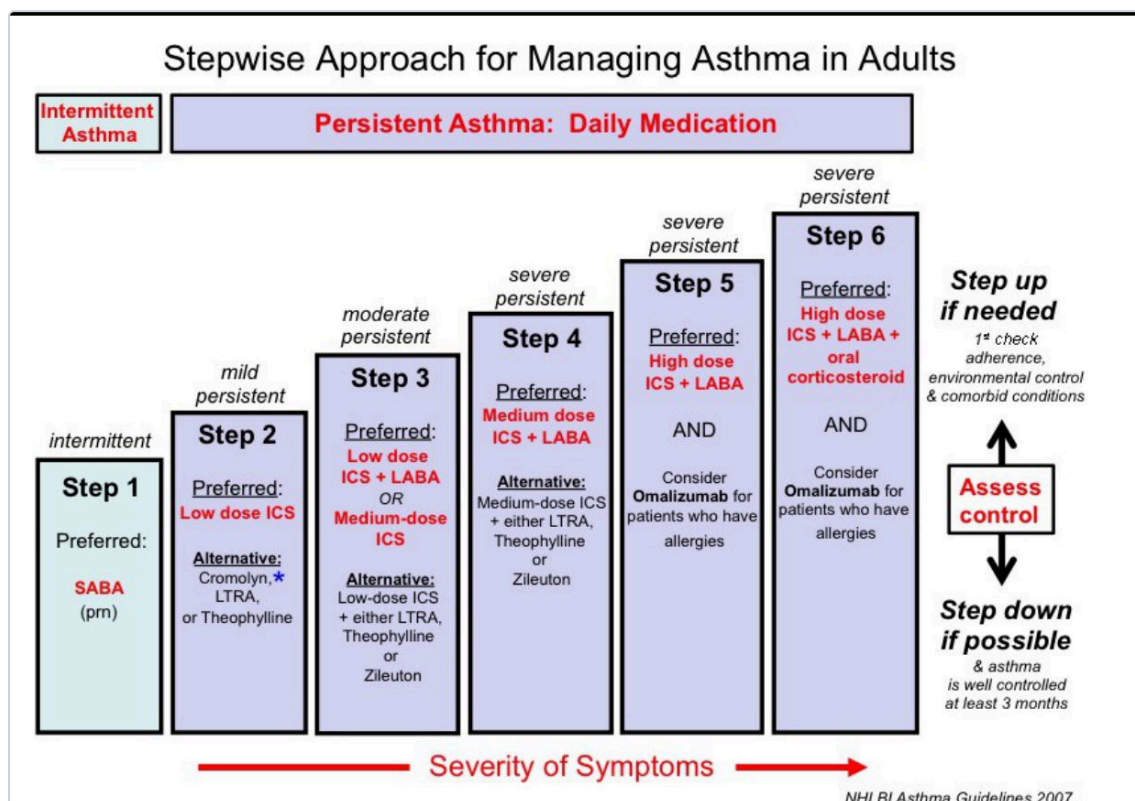
Classification	Daytime Symptoms	Nighttime Symptoms	PEF or FEV1
Mild intermittent	≤2 per week	<2 per month	≥80%
Mild persistent	>2 per week, but <1 per day	>2 per month	≥80%
Moderate persistent	Daily symptoms	>1 per week	60%-80%
Severe persistent	Continued day symptoms	Frequent night symptoms	<60%

Slide 19: Asthma severity classification

☆☆☆ Stepwise approach, adults (Slide 20, NHLBI 2007):

Step	Severity	★ Preferred	Alternative
1	Intermittent	★ SABA prn	
2	Mild persistent	★ Low-dose ICS	Cromolyn, LTRA, or theophylline
3	Moderate persistent	★ Low-dose ICS + LABA or medium-dose ICS	Low-dose ICS + LTRA, theophylline or zileuton
4	Severe persistent	★ Medium-dose ICS + LABA	Medium-dose ICS + LTRA, theophylline or zileuton
5	Severe persistent	★ High-dose ICS + LABA, consider omalizumab (allergic)	
6	Severe persistent	★ High-dose ICS + LABA + oral corticosteroid, consider omalizumab	

- ★ **Step up if needed**, but **first check adherence, inhaler technique, environmental control, comorbidities**
- ★ **Step down if possible** once **well controlled for ≥ 3 months**
- SABA as needed at every step



Slide 20: Stepwise approach for managing asthma in adults

Drug classes (Slide 21): inhaled **β_2 -agonists** · **corticosteroids** (inhaled and oral) · **montelukast** (leukotriene modifier) · **theophylline** · ★ **omalizumab** (anti-IgE monoclonal antibody)

⚠ [added: current GINA guidance (2019+) advises against SABA-only treatment. Even mild asthma should get ICS-containing reliever therapy (e.g., as-needed low-dose ICS-formoterol). The lecture's own risk-factor slide lists "SABA-only treatment" as a cause of exacerbations. For this exam, answer by the lecture's NHLBI table, but know the update.]

🔄 **Recap: Stepwise.** Classify by day symptoms, night symptoms and FEV1 (≥ 80 / 60-80 / < 60). Step 1 SABA; 2 low ICS; 3 low ICS + LABA; 4 medium ICS + LABA; 5 high ICS + LABA ± omalizumab; 6 + oral steroid. Check adherence/technique before stepping up; step down after 3 months controlled.

🔗 **Link:** Even controlled asthma can flare acutely, and that's where patients die.

📌 **Up next: Acute exacerbation.**

2.8 Acute Asthma Exacerbation (Slides 22-27)

Definitions (Slide 22):

Term	Meaning
Asthma exacerbation	Usually reversible lower-airway obstruction (bronchospasm); symptoms worsen over a short time (acute/subacute) with a change from baseline lung function
Bronchospasm	Bronchial-muscle constriction causing obstruction within seconds to minutes ; can also be triggered by drugs or mechanical ventilation

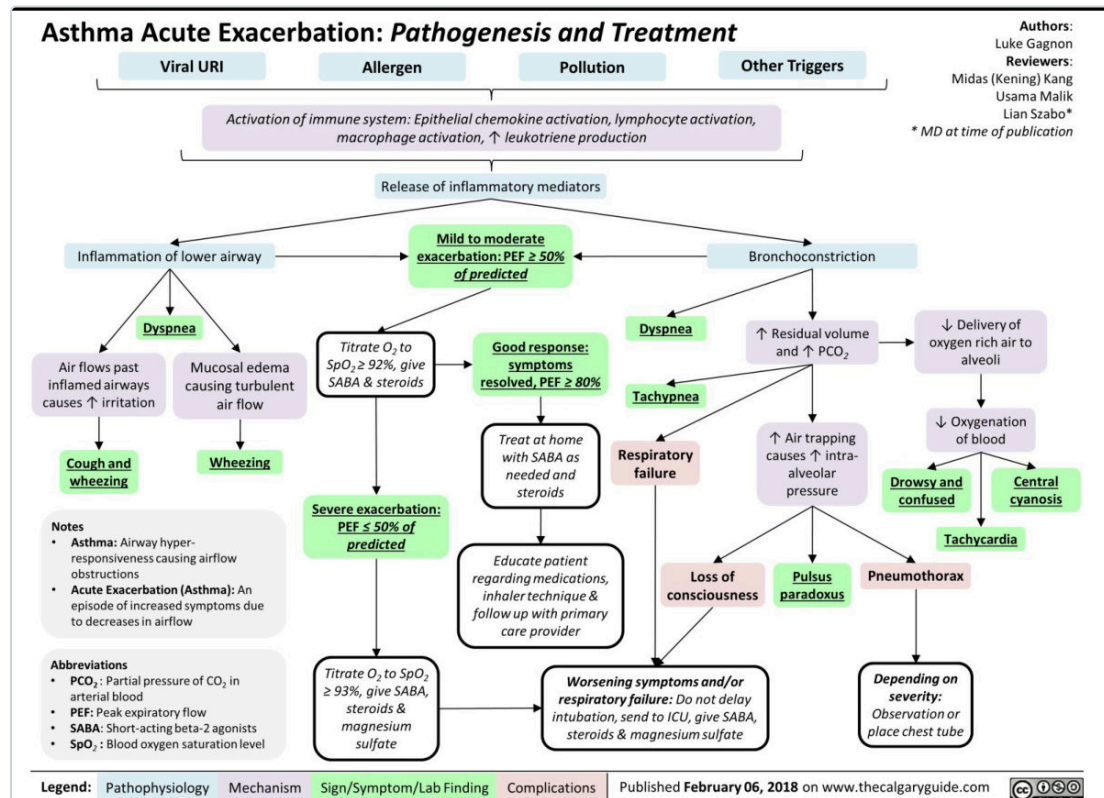
Term	Meaning
★ Status asthmaticus	Severe exacerbation that progresses rapidly and does not respond to standard therapy

★★ Risk factors for exacerbations (Slide 23):

- Trigger exposure
- ★ SABA-only treatment / SABA overuse
- ★ Inadequate ICS or poor adherence
- Incorrect inhaler technique
- ★ Prior intubation or ICU admission
- ≥ 1 severe exacerbation in the past year
- ★ FEV1 < 60% predicted
- High bronchodilator responsiveness
- Blood eosinophilia

Pathogenesis (Slide 24, Calgary Guide): triggers (viral URI, allergen, pollution, others) → immune activation ($\uparrow$ leukotrienes) → inflammatory mediators → **airway inflammation** (cough, wheeze, dyspnoea) + **bronchoconstriction** → $\uparrow$ RV and PCO₂, **air trapping** ($\uparrow$ intra-alveolar pressure → **pulsus paradoxus**, **pneumothorax**, loss of consciousness) → $\downarrow$ oxygenation (**drowsiness/confusion**, **central cyanosis**, **tachycardia**) → **respiratory failure**.

- **Mild-moderate:** PEF $\geq 50\%$ → titrate O₂ to SpO₂ $\geq 92\%$, **SABA + steroids** → good response (PEF $\geq 80\%$) → home with SABA prn + steroids + education
- **Severe:** PEF $\leq 50\%$ → O₂ to SpO₂ $\geq 93\%$, **SABA, steroids, magnesium sulfate**
- **Worsening / respiratory failure** → **don't delay intubation**, ICU, SABA, steroids, MgSO₄
- **Pneumothorax** → observe or chest tube



Slide 24: Acute exacerbation: pathogenesis and treatment (Calgary Guide)

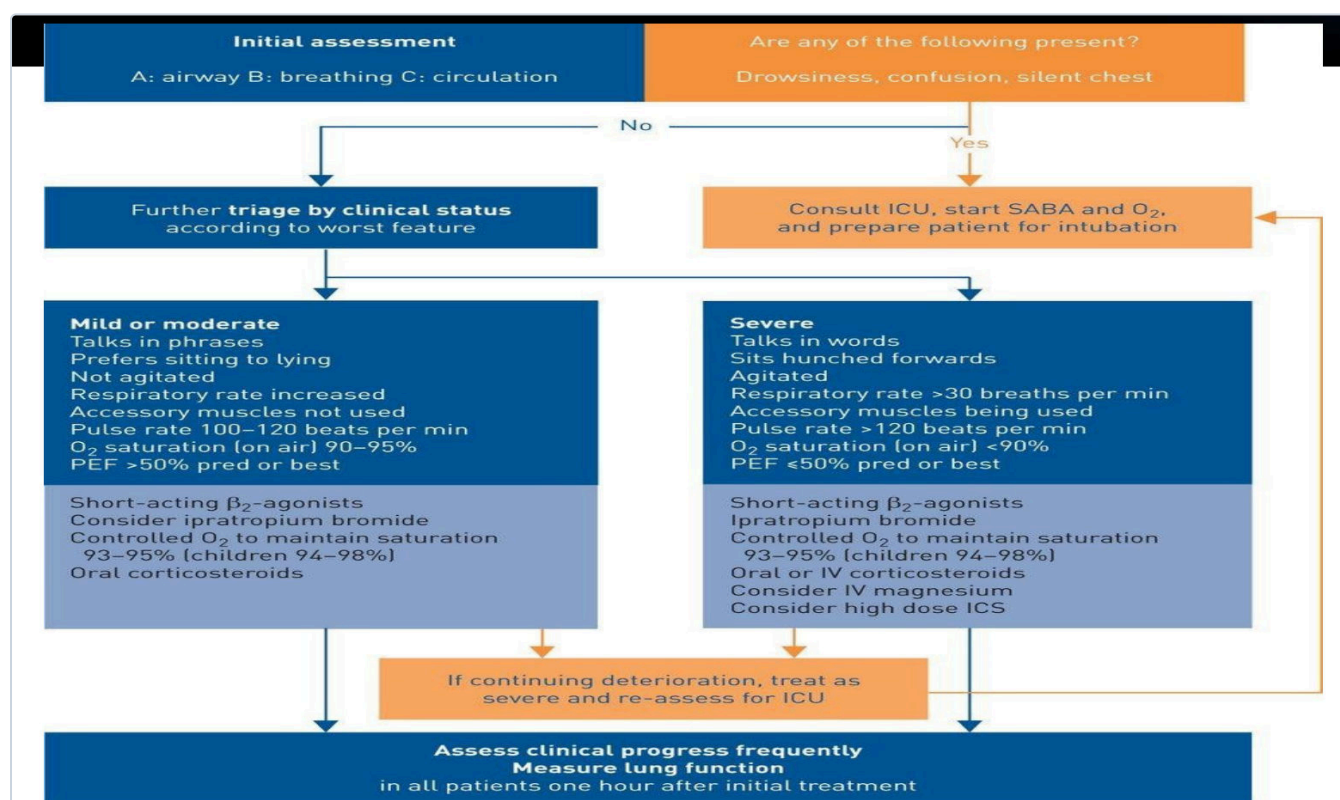
Clinical features (Slide 25): dyspnoea, $\uparrow$ cough · tachypnoea, tachycardia, ★ **pulsus paradoxus**, ★ **hypoxaemia** (SpO₂ < 90%, possible cyanosis), ★ **altered mental status** · prolonged

expiration, wheeze, ★ **silent chest**, hyperresonance, **low, poorly moving diaphragm** · **accessory muscles**

★★★ **Triage by worst feature (Slide 27):**

	Mild / moderate	★ Severe
Speech	Phrases	★ Words
Posture	Prefers sitting	Sits hunched forward
Agitation	Not agitated	Agitated
RR	Increased	★ > 30/min
Accessory muscles	Not used	★ Used
Pulse	100-120	★ > 120
SpO ₂ (air)	90-95%	★ < 90%
PEF	> 50%	★ ≤ 50%
Treatment	SABA , consider ipratropium, O ₂ to 93-95% (children 94-98%), oral steroids	SABA + ipratropium , O ₂ to 93-95%, oral or IV steroids , ★ consider IV magnesium , consider high-dose ICS

★ **Red flags first (Slide 27):** ★ **drowsiness, confusion, silent chest** → **consult ICU, start SABA + O₂, prepare for intubation**. ★ If deteriorating → treat as severe and reassess for ICU. ★ **Reassess lung function 1 hour after initial treatment** in all patients.



Slide 27: Acute exacerbation triage and management algorithm

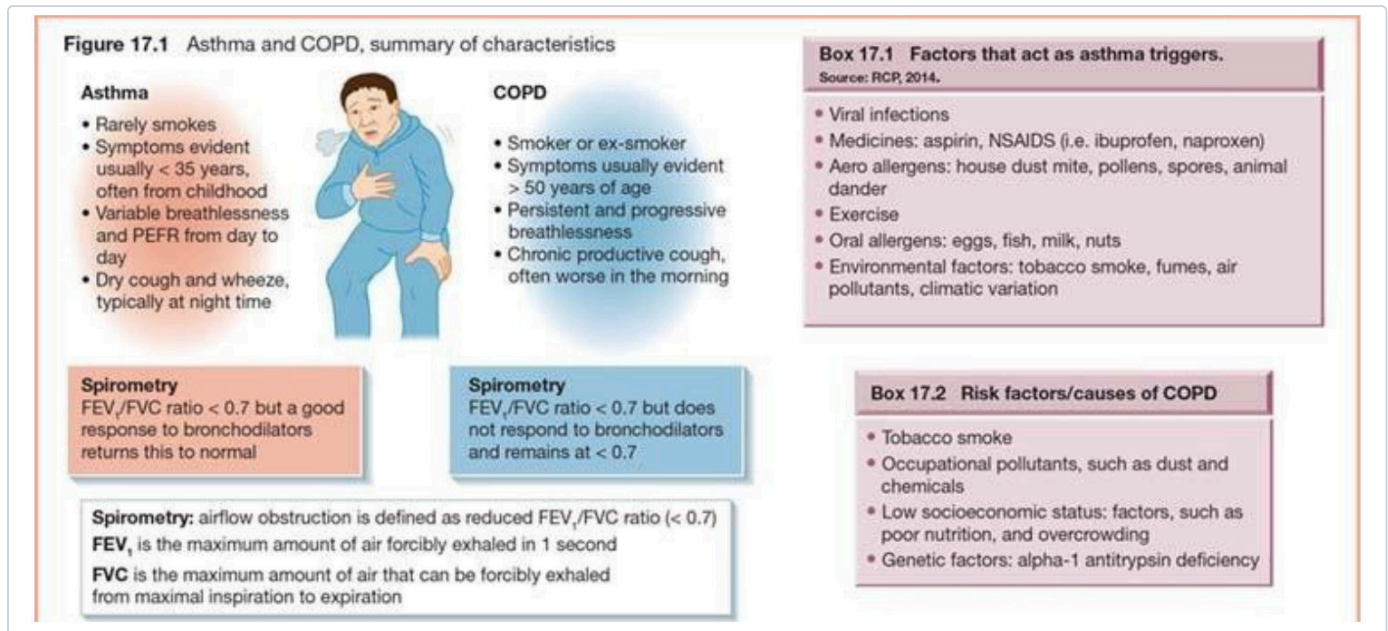
★★ **Severe / life-threatening management (Slide 26):**

1. **Urgent respiratory support:** O₂; **prepare for intubation and ventilation**; watch for impending respiratory failure
2. **Maximise drugs:** ★ **nebulised SABA + nebulised SAMA + oral or IV glucocorticoids**; ★ **consider single dose IV magnesium sulfate**; consider ICS

3. Fluids (preferably oral) in infants and young children

🔄 **Recap: Acute.** Severe = words only, RR > 30, HR > 120, SpO₂ < 90%, PEF ≤ 50%. Treat with nebulised SABA + ipratropium, steroids, O₂ 93–95%, IV Mg if severe. Drowsy, confused or silent chest → ICU and intubate. Reassess at 1 hour.

2.9 Summary (Slide 28)



Slide 28: Asthma vs COPD summary; asthma triggers; COPD risk factors

★ **Asthma triggers (Box 17.1):** viral infections · aspirin/NSAIDs (ibuprofen, naproxen) · **aero-allergens** (house-dust mite, pollen, spores, animal dander) · **exercise** · **oral allergens** (eggs, fish, milk, nuts) · environment (**tobacco smoke**, fumes, pollutants, **climate change**)

3. High-Yield Comparison Tables

Table A: Asthma vs COPD (see the added diagram at the top)

Table B: Allergic vs non-allergic (see §2.2)

Table C: Severity classification (see §2.7)

Table D: Stepwise therapy (see §2.7)

Table E: Mild/moderate vs severe exacerbation (see §2.8)

Table F: Diagnostic thresholds

Test	Asthma threshold
FEV ₁ /FVC	< 70% (during obstruction)
Bronchodilator reversibility	★ > 12% ↑ FEV ₁
Methacholine/histamine challenge	★ > 20% ↓ FEV ₁

Test	Asthma threshold
PEF normal	~400–600 L/min
Severe exacerbation PEF	≤ 50%
Good response	PEF ≥ 80%

4. Rules, Exceptions, Patterns

- **Reversible = asthma; irreversible = COPD.**
- **Reversibility > 12%; provocation fall > 20%.**
- **Allergic asthma responds to ICS; non-allergic/adult-onset responds poorly.**
- **Aspirin/NSAIDs mostly provoke non-allergic (intrinsic) asthma.**
- **CXR is normal in mild-moderate asthma.**
- **In an attack, low PaCO₂ is expected; normal/high PaCO₂ = tiring = danger.**
- **Silent chest is worse than wheeze:** no air is moving.
- **Check adherence and inhaler technique before stepping up.**
- **Step down only after ≥ 3 months of control.**
- **Omalizumab (anti-IgE) for allergic severe asthma (steps 5-6).**
- **Exercise-induced: SABA 15-30 min before; daily ICS if ≥ 4×/week.**
- **O₂ target: asthma 93-95% (children 94-98%); COPD 88-92%.**

5. Clinical Correlations

- **Child with eczema + allergic rhinitis + nocturnal cough/wheeze** → allergic asthma (atopic march *[added]*).
- **Middle-aged adult with nasal polyps and attacks after ibuprofen** → aspirin-exacerbated respiratory disease *[polyps added]*.
- **Chronic dry cough only, normal exam** → cough-variant asthma → methacholine challenge.
- **New mould or damp housing** → environmental trigger (Case 2).
- **β-blockers** can precipitate bronchospasm in asthmatics *[added]*.
- **Sudden deterioration with unilateral absent breath sounds in an attack** → pneumothorax (air trapping, Slide 24).
- **Over-use of SABA (> 2 canisters/month *[added]*)** = poor control, high exacerbation risk.

6. Rapid Revision (Last-Day)

- Asthma = chronic inflammation + **hyperresponsiveness** + **reversible** obstruction.
- Allergic: childhood, **extrinsic**, atopy, **↑ IgE**, type I, better course. Non-allergic: > 40 y, **intrinsic**, normal IgE, **aspirin/NSAIDs**, worse course.
- Risks: family history, allergy, atopy, pollution (cooking/heating).
- **Th2-driven:** hyperresponsiveness · eosinophilic inflammation (oedema → collapse) · obstruction (vagal bronchospasm, mucus, goblet-cell hyperplasia).
- Symptoms: episodic dyspnoea, **night-time dry cough**, chest tightness, wheeze; atopy history.
- Signs: wheeze, prolonged expiration, **silent chest**, hyperresonance, **pulsus paradoxus**, cyanosis.
- Subtypes: allergic (ICS works), non-allergic/adult-onset (ICS poor), **cough-variant**, aspirin-exacerbated, overlap, occupational, **exercise-induced** (15 min / 1 h; SABA 15–30 min before; ICS if ≥ 4×/wk).

- Spirometry: **FEV1/FVC < 70%, > 12% reversibility. Methacholine: > 20% fall.** PEF 400–600 L/min.
- Eosinophilia, ↑ IgE, skin prick; **Curschmann spirals, Charcot-Leyden crystals.** CXR normal (mild-moderate). **Normal/↑ PaCO₂ = bad.**
- Classification: intermittent (≤ 2/wk, < 2/mo night, ≥ 80%) · mild persistent (> 2/wk, > 2/mo, ≥ 80%) · moderate (daily, > 1/wk, 60–80%) · severe (continuous, frequent, < 60%).
- Steps: **1 SABA · 2 low ICS · 3 low ICS + LABA · 4 med ICS + LABA · 5 high ICS + LABA ± omalizumab · 6 + oral steroid.**
- Exacerbation risk: SABA-only/overuse, poor ICS adherence, bad technique, **prior ICU/intubation**, recent severe attack, **FEV1 < 60%**, eosinophilia.
- Severe attack: **words, RR > 30, HR > 120, SpO₂ < 90%, PEF ≤ 50%.** Rx: **nebulised SABA + ipratropium, steroids, O₂ 93–95%, IV MgSO₄.**
- **Drowsy/confused/silent chest → ICU, intubate.** Reassess at 1 h.
- **Status asthmaticus** = severe, rapidly progressive, unresponsive to standard therapy.

7. Exam-Style Questions

Q1 (MCQ, lecture Case 1, Slide 29). 20 ♂, 4 months of episodic dyspnoea and dry cough, **2 episodes/week**, resolving with rest; **wakes 2×/month**. Exam normal, SpO₂ 98%. **FEV1/FVC 0.85, FEV1 81%.** Best initial pharmacotherapy? **a. Albuterol inhaler** b. Fluticasone c. Montelukast d. Cromolyn e. Mometasone + zafirlukast f. Salmeterol **Answer: a** (per the lecture's NHLBI table). *[reasoning added; no key on the slide]* Daytime ≤ 2/week, night-time ≈ 2/month, FEV1 ≥ 80% → **mild intermittent → step 1: SABA prn.** ⚠ Night waking "twice a month" sits on the boundary (< 2 vs > 2/month), and current GINA would advise as-needed **ICS-formoterol** rather than SABA alone. Salmeterol (LABA) must **never be used alone** in asthma *[added]*.

Q2 (MCQ, lecture Case 2, Slide 30). 6-year-old with eczema, 2 months of cough, nasal congestion and **intermittent wheeze**; family moved into new affordable housing 3 months ago; scattered wheezes. Next step? a) B-cell flow cytometry **b) Spirometry** c) Skin prick testing d) Throat culture e) IgA levels f) DHR test **Answer: b.** *[reasoning added]* Atopy (eczema) + recurrent wheeze + a possible new environmental trigger → suspect asthma → **confirm with spirometry and reversibility** (feasible from about age 5). Skin prick testing identifies triggers after diagnosis. The immunodeficiency tests (flow cytometry, IgA, DHR for CGD) don't fit 3 self-limited URTIs and one otitis media.

Q3 (MCQ, lecture Case 3, Slide 31). 6-year-old asthmatic, missed medicines; RR 40, retractions, wheeze; given nebulised albuterol + ipratropium and IV methylprednisolone. One hour later: **limp, lethargic**; MgSO₄ given; **HR 150, RR falls to 22, no wheeze heard.** Next step? **b) Intubate with mechanical ventilation** **Answer: b.** *[reasoning added]* **Lethargy + "silent chest" + falling RR = exhaustion and impending respiratory failure**, not improvement (Slide 27: drowsiness, confusion, silent chest → prepare for intubation). NIPPV needs a cooperative, alert patient. Needle thoracostomy needs signs of tension pneumothorax (unilateral findings, hypotension). More drugs waste time.

Q4 (MCQ). Which ABG during an acute attack is most worrying? a) pH 7.48, PaCO₂ 30 **b) pH 7.38, PaCO₂ 42** c) pH 7.46, PaCO₂ 33 d) pH 7.50, PaCO₂ 28 **Answer: b.** "Normal" CO₂ in an attack means the patient is tiring.

Q5 (Short answer). A patient has daily symptoms, wakes > 1 night/week, FEV1 70%. Classify, and give the preferred treatment step. **Answer: Moderate persistent → step 3: low-dose ICS + LABA or medium-dose ICS** (+ SABA prn). Check adherence and technique first; review and step down after 3 months of control.

8. Exam Pitfalls & Traps

1. **Asthma = irreversible obstruction.** No: **reversible** ($> 12\%$ FEV1).
2. **Methacholine threshold:** a **$> 20\%$ fall** in FEV1, not 12%.
3. **Normal PaCO₂ in an attack is reassuring.** It's a **danger sign**.
4. **Silent chest = improving.** It's **life-threatening** (no airflow).
5. **LABA alone** in asthma. Always with ICS [*added safety point*].
6. **Non-allergic asthma responds well to ICS.** It responds **poorly**; allergic responds well.
7. **High IgE in intrinsic asthma.** Total IgE is **normal** in non-allergic.
8. **Stepping up without checking adherence/technique.** The slide says check these **first**.
9. **O₂ target confusion:** asthma **93-95%** (children 94-98%); COPD **88-92%**.
10. **CXR to diagnose asthma.** It's normal in mild-moderate; used in severe attacks to exclude other causes.
11. ⚠ **Guideline age:** the stepwise slide is **NHLBI 2007** (SABA alone at step 1). The lecture's own risk-factor slide calls SABA-only treatment a risk. Current GINA avoids SABA-only. Answer per the lecture unless asked about current guidance.
12. ⚠ **Slide numbers vs Calgary figure:** the triage slide targets O₂ **93-95%**, while the Calgary figure says titrate to SpO₂ $\geq 92\%$ (mild) / $\geq 93\%$ (severe). They're close; quote 93-95% for adults.